Mohammad Omar Nurshad

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Research profile

Statistics graduate (University of Chittagong) working on climate-sensitive disease modelling, count-data regression and survey epidemiology. Seeking fully funded MSc or PhD study in biostatistics, epidemiology and health data science. Interests: Applied statistics, biostatistics, epidemiology, climate and health, statistical learning.

Education

M.S. in Statistics, University of Chittagong, Bangladesh. Thesis submitted September 2026. CGPA 3.25/4.00 (M.S. examination 2024, held May–July 2026; result published 14 October 2026). Thesis and thesis defence graded A+.

B.Sc. (Honours) in Statistics, University of Chittagong, 2020–2025. CGPA 3.59/4.00 (160 credits; transcript published 8 September 2025; provisional certificate 13 January 2026). Medium of instruction: English (Registrar certificate, 19 January 2026).

Master's thesis

"Modeling and Predicting Daily Dengue Cases in Bangladesh Using Meteorological Factors: A Comparative Study of Statistical and Machine-Learning Approaches." Department of Statistics, University of Chittagong, September 2026. Supervisor: Prof. Mohammad Salim Zahangir.

- Data: DGHS daily dengue counts linked to BMD weather records for all eight divisions. Track A (association) uses 2022–2024 division-days; Track B (forecasting) is tested on 2025.
- Association: negative binomial (NB2) models chosen after overdispersion diagnostics (Pearson dispersion 30.45 to about 1.14); lag comparison (28-day lag lowest AIC, 14-day close); sensitivity with ZINB, GEE and PPML. Temperature was the one robust association (nonlinear, delayed, ecological, not causal).
- Forecasting: leakage-controlled 7-day-ahead models (persistence, log-ridge, random forest, XGBoost) with rolling-origin validation. 2025 division-day MAE: log-ridge 8.854, persistence 9.451, XGBoost 9.792, random forest 11.144. Peaks were underpredicted (P95+ MAE 84.86).
- Limitations: ecological design; temporal rather than prospective test; incomplete upstream data provenance.

Publications and preprints

Faisal MA, **Nurshad MO**, Alam MT, Barua A. Sedentary Behavior and Eating Habits Associated with Fast Food Consumption Among Bangladeshi University Students: The Moderating Role of Residential Status. **medRxiv preprint**, posted 28 April 2026, not certified by peer review. doi:10.64898/2026.04.24.26351719. Reported as under revision at PLOS ONE (PONE-D-26-19727).

My role: conceptualization, investigation, statistical analysis, data curation, writing (original draft), writing (review and editing). Cross-sectional survey of 433 students; proportional-odds ordinal regression with residential-status interactions, Brant test, chi-square and Fisher's exact tests.

No peer-reviewed publication is claimed.

Conference

Author, Life and Health Science theme, Scholars' Convergence: Research & Innovation Summit 2026, Chittagong University Scientific Society, 31 January 2026 (certificate of participation).

Awards and funding

Nominated for the Information and Communication Technology (ICT) Division, Government of Bangladesh, fellowship for master's-level research in the IT sector, fiscal year 2026–27 (one of 31 nominees; official notification dated 13 October 2026). Application proposal: "Development of Explainable Statistical and Machine-Learning Models for Forecasting Daily Reported Dengue Cases Using Meteorological Factors in Bangladesh."

Training

- Data Analytics Career Bootcamp (16 weeks), Human Development Network Bangladesh; completed 3 December 2025; certificate issued 29 January 2026; 76/100.
- Introduction to Statistical Thinking for Non-Statistical Audiences, ResearchOrbit, 3 October 2025.
- Data and Decision Science: Higher Studies, Perspectives, Careers, and Research Avenues, ResearchOrbit, 27 September 2025.

- Digital Marketing, E-commerce Management and Search Engine Optimization (216 hours), Sheikh Kamal IT Training and Incubation Center Project, Bangladesh Hi-Tech Park Authority.

Methods and software

Demonstrated in research: negative binomial, Poisson, PPML and GEE count models; ordinal logistic regression; chi-square and Fisher's exact tests; ridge regression, random forest, XGBoost; rolling-origin temporal validation.
Software: SPSS v25 and R 4.1.3 (preprint). Working knowledge of Stata, SAS and Python (coursework and training).
Developing: Bayesian and spatiotemporal modelling, probabilistic forecasting, causal inference.

Experience and activities

- Academic tutor, Mathematics, Statistics and ICT (self-employed), February 2021–January 2024
- Research assistant, Creative Scholar BD, 2023–2025
- Committee member, Programming Club, University of Chittagong; member, CURHS research group

References

Prof. Mohammad Salim Zahangir, Ph.D., Professor, Department of Statistics, University of Chittagong (research supervisor). salim.zahangir@cu.ac.bd

Prof. Dr. Soma Chowdhury Biswas, Professor of Statistics, Department of Statistics, University of Chittagong. soma.stat@cu.ac.bd